Check for updates

Original Manuscript

Hammill Institute on Disabilities

Focus on Autism and Other Developmental Disabilities
1–13
© Hammill Institute on Disabilities 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10883576241232894
focus.sagepub.com

Sage

Effects of Embedding Special Interest Area in Instruction on the Engagement and Out-of-Seat Behaviors of Children With Autism Spectrum Disorder

Gospel Y. Kim, PhD, BCBA-D[1], SoHyun Lee, PhD[2], Kathleen N. Tuck, PhD, BCBA-D[3], and Jose R. Martinez, PhD, BCBA-D[4]

Abstract

The purpose of this study was to examine the effects of embedding special interest areas (SIAs) in whole-group (WG) instruction on engagement and out-of-seat (OOS) behaviors for young children with autism spectrum disorder (ASD) in general education classrooms. A multiple baseline across participants design was used to evaluate the functional relation between the SIA-embedded instruction and engagement and OOS behaviors of all three children. Non-targeted probes were conducted after every intervention session to investigate whether there was behavioral change when the SIAs were removed from instruction. Results indicated that embedding SIAs in WG instruction may increase the engagement behavior and decrease the OOS behavior of children with ASD; behavioral changes were maintained over time for all children.

Keywords

out-of-seat behavior, special interest areas, behavior change

Challenging behavior (CB) is defined as a repeated pattern of behavior that may interfere with a child's engagement in interactions with peers or adults, including aggression, elopement, and destroying objects (Smith & Fox, 2003). Approximately 5% to 25% of young children exhibit CB (Hemmeter & Conroy, 2018). Persistent CB can interfere with access to instructional opportunities and social-emotional development (Gilliam, 2005). Without early intervention, the rate and severity of CBs are likely to increase and, in turn, have a negative impact on the long-term outcomes in home, school, and community settings (Ogundele, 2018).

Challenging behavior can negatively impact children's performance in school by reducing access to instructional activities. Socially mediated CBs such as tantrums and out-of-seat (OOS) behaviors are common in preschools (Kiely et al., 2016). While these behaviors may be expected from children as they acquire self-regulation and language, leaving instructional areas may lead to decreased time engaging in effective instruction and increased placement where teaching is limited (Morris & Horner, 2016; Smith & Fox, 2003). Moreover, OOS behavior has been identified as a barrier to the inclusion of children with autism spectrum disorder (ASD; Crosland & Dunlap, 2012). One effective strategy to reduce OOS behavior is to modify instructional methods or approaches to increase the likelihood of children engaging in instructional activities (McComas et al., 2000).

Engagement

Engagement is defined as actively attending and participating in classroom activities by exhibiting behaviors such as visual fixation, manipulation of materials, vocalization, or approach during ongoing activities (Ridley et al., 2000). Engagement mediates the effects of early childhood education on developmental outcomes (McWilliam et al., 1985; Raspa et al., 2001) because frequent positive engagement leads to improved social skills, cognitive development, and communication skills (Willford et al., 2013). Thus, engagement can be considered a pivotal behavior through which children can acquire other skills across developmental

[1]Juniper Gardens Children's Project, Kansas City, KS, USA
[2]Ewha Womans University, Seoul, South Korea
[3]The University of Kansas, Lawrence, USA
[4]Florida Autism Center, Daytona Beach, USA

Corresponding Author:
Gospel Y. Kim, UC Davis MIND Institute, 2825 50th St, Sacramento, CA 95817, USA.
Email: goykim@ucdavis.edu

domains. Research on engagement suggests that children are less engaged in larger groups, and whole-group (WG) instruction is widely used in early childhood education classrooms (Powell et al., 2008). Supporting engagement for all children including children with ASD is important given they display lower levels of engagement than their peers (Keen, 2009; Kishida & Kemp, 2006).

Researchers have investigated strategies to improve the engagement of young children with ASD (e.g., Rogers & Dawson, 2020). Antecedent strategies such as visual supports (Steinbrenner et al., 2020), structured environmental arrangement (Bennett et al., 2011), curricular revision (Fleury & Schwartz, 2017), and choice-making (Rispoli et al., 2013) have all been demonstrated to be effective at improving the engagement of young children with ASD in instructional settings. Consequence strategies such as differential reinforcement (Fisher et al., 2014) and extinction procedures which include not responding to a child's inappropriate initiations are also effective at increasing engagement (Kodak et al., 2012). Despite such extensive evidence for using the interventions supporting engagement and CB, few studies have explored inclusive, WG settings (Hemmeter et al., 2011; Zimmerman et al., 2019). Given the frequency of the WG instruction, it is important to incorporate effective strategies promoting engagement within this instructional context (DiCarlo et al., 2012).

Improving the engagement of children with ASD exhibiting CBs (e.g., OOS behavior) may require individualized interventions. Research suggests that disability characteristics may impact engagement and development (Adamson et al., 2004). For example, repetitive or restricted behavior and interests of ASD (American Psychiatric Association, 2013) may negatively impact engagement and development if not addressed. Such behaviors include stereotypical motor movements or vocalizations, insistence on sameness, sensory activities, and restricted interests (i.e., special interest areas; SIAs). Special interest areas are often expressed as restricted or repetitive use of an object, preoccupations with particular stimuli, or fascinations for specific topics or activities (Bishop et al., 2013). Utilizing the SIAs of children with ASD as a strengths-based approach to intervention may increase engagement and reduce CB (Boyd et al., 2011).

Special Interest Areas

Special interest areas have been used in educational practice as both consequence and antecedent-based interventions (Gunn & Delafield-Butt, 2016). The consequence-based approach provides items of interest contingent on desired behavior, whereas the antecedent-based approach embeds SIAs in tasks or activities to increase the motivation to engage in appropriate behaviors (Gunn & Delafield-Butt, 2016). Special interest areas can be used to improve social behaviors (Boyd et al., 2007; Koegel et al., 2013), academic skills (El Zein et al., 2014), and CB (Mancil & Pearl, 2008). Despite the use of SIAs in different settings and contexts for students of various ages, to our knowledge, no study exists examining the effectiveness of SIAs on engagement when general educators in inclusive settings were the implementers (Ninci et al., 2020).

The purpose of this study was to examine the effects of embedding SIAs on engagement and OOS behaviors of children with ASD during WG instruction in general education preschool classrooms. The following research questions were addressed in this study: Does embedding SIAs in WG instruction increase the engagement of children with ASD? Does embedding SIAs in WG instruction decrease the OOS behaviors of children with ASD? Do increases in engagement and decreases in OOS behaviors maintain over time when SIAs are removed from instructional activities? Do changes in engagement and OOS behaviors during non-targeted WG instruction match the levels displayed during targeted WG instruction when SIAs are present? What are the general educators' perspectives on using SIA-embedded instruction?

Method

Participants and Implementers

Three children with ASD participated in this study. Inclusion criteria for the participants were (a) age range between 3 years 0 months and 5 years 11 months; (b) clinical diagnosis of ASD; (c) enrolled in an inclusive classroom; (d) have an identifiable SIA; (e) demonstrate low levels of engagement in WG instruction; (f) demonstrate a high frequency of OOS behavior in WG instruction; (g) regular school attendance; and (h) consent to participate in the study. Participant information is found in Table 1.

Joy was a 5-year-old Asian female diagnosed with ASD who used one-word utterances and picture symbol cards to augment her receptive and expressive communication. Her mother and teachers reported her engagement in restricted and repetitive behaviors frequently across daily routines and her high levels of self-stimulatory behaviors during wait time or while maintaining her seat to listen to others. She frequently laughed in a loud voice out of context, left her seat, and played with toys during WG instruction (e.g., storytelling). Evan was a 5-year-old Asian male diagnosed with ASD who communicated verbally using three-word sentences and responded to questions using echolalia. During the study, he continued the speech and language therapy that he had been receiving for 1 year. Evan's mother and his teachers reported his restricted and repetitive motor movements (e.g., combined movements of hand flapping and touching his hands on the floor) when he was asked to wait and keep his seat, his high frequency of staring at the doors or seeking other play areas during WG instruction, not responding when his name is called to ask for

Table 1. Information on Participants in the Study of Effects of Special Interest Areas on Engagement and Out-of-Seat Behaviors.

Characteristic	Joy	Evan	Liam
Age (years)	5	5	3
Diagnosis	ASD	ASD	ASD
Characteristics related to target behaviors	Demonstrates self-stimulating behavior (e.g., repetitively feeling texture of clothes) during times required to wait for a turn or listen to others. Suddenly laughs in a loud voice out of context & attracts attention from peers & teachers. Leaves her seat and plays with toys during static whole-group activities (e.g., storytelling, book reading).	Stares at the doors or other play areas during whole-group activities. Does not respond when his name is called to give tasks. Shakes hand & hits the floor with hand repetitively when asked to imitate movements. Frequently runs out of the classroom to press the elevator buttons during indoor activities.	Counts the numbers on objects such as clocks or calendars repeatedly during activities. Physical prompts from peers & teachers are required to participate in activities with rules (e.g., physical game activity). Remains in his seat less than 3 min during static whole-group activities (e.g., storytelling, book reading).

Note. ASD = autism spectrum disorder.

participation, and running out of the classroom to press the elevator buttons. Liam was a 3-year-old Asian male diagnosed with ASD who used a one-word utterance to verbally name the object/subject. His mother and teachers reported that he frequently engaged in behaviors or activities related to numbers, such as counting numbers on objects, lining up number toys, and watching videos with numbers. During classroom activities, teachers reported his fixation of interest in numbers that can be triggered within his physical environment (e.g., calendar) impedes his participation in activities. They also reported that he remained in his seat for less than 3 min during WG instruction that did not require tasks involving movements (e.g., book reading) and often needed physical prompts to follow the rules.

Three general educators (i.e., each child's general educator) served as the intervention agents in this study. All three educators had bachelor's degrees in early childhood education and received teacher licensure to work in public schools. Two educators had 2 years of experience working in an inclusive classroom, and the other educator had no experience related to inclusion.

Setting and Materials

The study was conducted in three general education classrooms in a public preschool in Seoul, South Korea. Joy and Evan's classroom consisted of 22 children of ages 4–5 years. Liam's classroom consisted of 16 children of ages 3–4 years. Each classroom had one lead general educator, a special educator, and a paraeducator. During the study, however, only the general educator was present in the classrooms across all sessions. The sessions took place in the circle time area where two 2 m × 3 m carpets were aligned together and surrounded by chairs. The WG instruction took 20 to 30 min and occurred once in the morning and the afternoon. The general educator began WG with a greeting to gain children's attention through a short song, the introduction of the main activity (e.g., gameplay, book reading), completion of the activity (e.g., tossing a ball while reviewing peer names and gaining peer attention; singing a song with hand motions), and reflection on the activity (e.g., children would be asked if they liked it, feedback on performance and social interactions would be provided). Each WG followed this schedule strictly with the nationwide curriculum assigned (Korea Ministry of Education, 2013).

Materials included the activity plan and the instructional materials that incorporated the SIAs identified for each child. The SIA materials were a toy microphone and car for Joy, elevators and counting floor numbers for Evan, and trains and counting train cars for Liam. While incorporating an actual car, elevator, or train was not possible in the classroom, objects representing the SIAs were added to ongoing instructional materials (e.g., using a microphone during the greeting for Joy and adding trains to the material tossed during gameplay for Liam). Special interest areas were sometimes incorporated as objects (e.g., a microphone), and at other times, these topics were included in book reading (e.g., reading a book with an elevator). The WG instructional plans that were modified using the SIAs of each child were scripted plans required for use as part of the national early childhood educational curriculum booklet. The lessons were selected for modification to meet the standards of compulsory education for preschoolers in Korea.

Response Definitions and Measurement System

The dependent variables in this study were engagement and OOS behaviors (adapted from Chazin et al., 2018). Engagement behavior is defined as the child oriented toward appropriate WG activity such as responding or looking at the speaker (e.g., teacher, peers) or interacting with instructional materials. Out-of-seat behavior is defined as

Table 2. Target Behavior Definitions.

Behavior	Definition	Examples	Non-examples
Engagement	Child follows directions in teacher-directed instruction; child demonstrates behaviors related to the suggested activity; child looks at the person instructing the activity; child orients the body toward the person leading the activity	Child raises hand to have turns; child responds appropriately to peers or teacher; child sings/plays instruments along with music; child looks at peers modeling for the whole group	Child raises hand repeatedly out of context; child sings/plays instruments/moves body repetitively and constantly after teacher has asked the whole group to stop
Out of seat	Child gets out of seat during the static activities (e.g., storytelling, book reading); child elopes from the activity area during dynamic activities (e.g., physical game activity, body movement)	Child lays down on carpet; child gets out of designated seat to move to another place; child orients body to non-targeted material by repositioning seat; child runs out of classroom.	Child gets out of seat to comply to the teacher's request; child is pushed by or bumped by others; child demonstrates repetitive behaviors (e.g., jumping up and down on the same spot) for self-regulation

the child discontinuing to maintain contact with the seat portion of either chair or carpet or eloping from the activity area during non-seated WG activities (e.g., physical game activity). Engagement and OOS behaviors are incompatible except for activities that occur when children are not seated (e.g., body movement activity); we included that as non-examples for OOS behavior (see Table 2). The operational definition, examples, and non-examples are shown in Table 2. The primary dependent variable was engagement, and the OOS behavior was the secondary. Examples of both behaviors were tailored to each activity presented during WG instruction.

Data for both target behaviors were collected via video recorded in every session. The camera was positioned by the corner of the classroom next to the backdoor out of children's sight and reach. Duration of child engagement and frequency of OOS behaviors were estimated using partial interval recording with 10s intervals. The total percentage of the intervals with each target behavior was calculated by dividing the number of intervals in which each behavior occurred by the total number of intervals observed, multiplied by 100.

Experimental Design and Visual Analysis

A multiple baseline across participants design (Ledford & Gast, 2018) was used to assess the presence of functional relation between the presence of SIA embedded in WG instruction and engagement and OOS behaviors of children with ASD. Condition changes were determined based on data stability (Ledford & Gast, 2018). The intervention was introduced to the first tier when data were stable; the intervention was introduced to subsequent tiers after four data points were collected; and data were stable with a counter-treatment trend. Visual analysis was used to assess the level, trend, and variability within and across conditions (Gast & Spriggs, 2014). A vertical analysis was conducted to determine if experimental control was established and detect covariation, a potential threat to internal validity. Experimental control was established when the behavioral change occurred only when the intervention was introduced in each tier. A functional relation between SIA intervention and engagement or OOS behavior was established when the desired behavior changes in each dependent variable occurred only when intervention was introduced, and those changes were consistent and replicated across all tiers.

Procedures

SIA Determination and Target Activities Selection. Participants' SIAs were identified through (1) educators' observations of the children and (2) caregivers' reports. The educators' observations were conducted daily over 2 weeks. Materials or items children voluntarily sought and demonstrated persistent engagement for more than 10 min every day were recorded by the educators. A face-to-face interview to generate a list of potential SIAs and a survey identifying the rankings of those potential SIAs derived from the interview were conducted with the caregivers. Target activities were selected for each child by a collaborative discussion with all stakeholders in each classroom (i.e., general educator, special educator, and para educator) about what they perceive as challenging activities for each child's engagement and learning.

Baseline. The duration of baseline sessions was 10 min. The general educator in each classroom was asked to instruct the WG activity based on the pre-created activity plan provided in the national curriculum guidebook. No supplementary materials or additional strategies were implemented to help the children's engagement. When the child exhibited OOS behaviors, the educators responded as they typically would

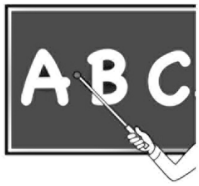
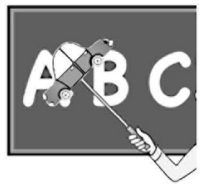
Name	SIA	Before Embedding SIA	Embedding Procedure	After Embedding SIA
Joy	Car; Toy; Microphone		• Cut visual materials in car shape • Used microphone as stimuli to share oral responses • Inserted image of a car or microphone on instructional materials	 Inserted image of a car on the pointer.
Evan	Numbers; Elevator		• Inserted image of numbers in typical activity materials • Used elevator and level of the floor as visual supports to show the sequence of the activity	 Inserted numbers to pictures in the book.
Liam	Numbers; Train		• Inserted image of numbers in typical activity materials • Placed visual materials in a train shaped frame • Used train as visual supports to understand the sequence of the activity	 Used train shape pictures on the Christmas decoration material.

Figure 1. Description of Special Interest Area (SIA) Embedding Procedure and Example
Note. Joy's instructional material adapted from materials on Openclipart. Creative Commons License CC0. SIA = special interest area.

using verbal and physical prompts, and these were consistently applied across all conditions for all three participants.

Modifying Activity Plan and Preparing Activity. The instructional plans were prepared daily, in accordance with weekly, monthly, and yearly education plans. For this study, the SIAs were embedded in the afternoon WG activities for 8 weeks, 3–5 sessions in a week. Each WG session consisted of five steps: transition in, introduction, execution, wrap-up, and transition out. The criteria for embedding SIAs into activities were (a) embedding SIAs across all five steps, (b) visually presenting the SIAs, and (c) introducing SIAs each time a new directive is given to the child (e.g., Evan's teacher may use the elevator to give the direction for next step by saying, "Now the elevator stopped at level 2, it's time to pass the ball to a friend behind you."). The following considerations were taken into account when modifying the activity plans: (a) the maintenance of the scope and sequence of the original activity plan, (b) the age-appropriateness of using SIAs, (c) caregivers' approval to use the SIA-embedded strategies, and (d) the possibility of expanding across behaviors and contexts. Examples of activity modifications using SIAs are in Figure 1. Based on the modified instructional plans, there were discussions among general and special educators on how to implement the intervention with fidelity by following the sequence of the activity and incorporating required materials with modifications for SIAs.

Instructing SIA-Embedded Activity. The general educators implemented the SIA-embedded activity 5 min after the beginning of each afternoon WG instruction and continued for a total duration of 10 min. The scope and sequence of the instruction were not different from how general educators typically delivered instruction, and there was no additional support from other staff or educators. The general educators were asked to embed SIAs throughout the activity as described in the modified instructional plans.

Probe on Non-Targeted Activity. During the morning WG instruction, the exploratory probe sessions were conducted to examine the effects of the intervention on engagement and OOS behaviors when SIAs were not embedded.

Maintenance. Three maintenance sessions were conducted for each child 3 weeks after the intervention was terminated. Similar to baseline sessions, SIAs were withdrawn from the activities during maintenance; sessions lasted 10 min.

Interobserver Agreement

Secondary observer data were collected in at least 25% (range = 25%–26.9%) of sessions across all participants for engagement and OOS behaviors to calculate interobserver agreement (IOA). Prior to observing and coding experimental sessions, a secondary data collector was trained on the data-collection procedures, which included reviewing the operational definitions of the target behaviors. Data for the reliability check were randomly selected from the videotaped recordings. The mean IOA for engagement behaviors was 99.3% (range = 95%–100%) for Joy, 99.1% (range = 91.6%–100%) for Evan, and 100% for Liam. The mean IOA for OOS was 100% for all participants.

Implementation Fidelity and Social Validity

Implementation fidelity data were collected in at least 35% (37.5%–39.1%) of randomly selected intervention sessions across participants. The first author used a checklist consisting of eight items regarding the accuracy and consistency of implementation by general educators (e.g., Did the general educator prepare the necessary instructional aids and materials according to the modified activity plan?). Special interest area–embedded activity plans and videos were examined to score either *yes* (1) or *no* (0) in response to each item. Implementation fidelity was calculated by converting the scores into a percentage, (total scored items/total number of items [8]) × 100. Mean fidelity was 91% (range = 70%–100%) for Joy's teacher, 93.8% (range = 70%–100%) for Evan's teacher, and 90.6% (range = 87.5%–100%) for Liam's teacher.

Social validity was measured at the end of the study using a 5-point Likert scale. The general educators were asked to complete questionnaires asking to what degree they agree or disagree on question items such as (a) embedding SIA in the instructional plan was easy, (b) implementing modified instruction was easy, (c) the increase on the child's engagement behavior was observed contingent upon the intervention, (d) the decrease on the child's OOS behavior was observed contingent upon the intervention, (e) using this strategy had a positive effect on classroom instruction in inclusive setting, (f) intention of using this strategy in the future, and (g) willingness to recommend this strategy to colleagues in inclusive settings. Additional descriptive open-ended questions on their views of the intervention were collected.

Results

Engagement

The percentage of intervals of engagement across each participant is shown in Figure 2. A functional relation was established between the implementation of the SIA intervention and the level of engagement exhibited by the children with ASD in the WG instruction. Engagement levels consistently increased across all participants when SIA intervention was implemented to levels higher than baseline; data patterns by the participants are described in the following sections.

Joy demonstrated a low level of engagement during baseline conditions, ranging from 5% to 12% of intervals. Her level of engagement increased immediately following the intervention onset and continued with an increasing trend that stabilized through Session 13. Engagement briefly decreased in Session 15 due to the participant's illness, then returned to high levels for the remainder of the intervention; around 55% of intervals engaged. Although the levels of engagement behavior displayed during non-targeted probe sessions did not match levels during SIA intervention, levels during the non-targeted probe sessions were higher than baseline levels when data were collected under the same conditions. Evan demonstrated a low level of engagement during baseline conditions (2%–17% of intervals). His engagement increased immediately following intervention onset and continued with an increasing trend that stabilized through Session 23 (around 60% of intervals engaged). Although levels of engagement displayed during the first six of non-targeted sessions did not match those during SIA intervention, levels of both conditions were similarly high from Session 14 to the last intervention session. Liam demonstrated a low level of engagement during baseline conditions (12%–30% of intervals). His engagement level immediately increased following intervention onset and continued with an increasing trend that stabilized through Session 21 (around 60% of intervals engaged). His level of engagement during non-targeted probes matched intervention levels. Across all children, data in both SIA-embedded and non-targeted conditions had an increasing trend (55%–60% of intervals) that was maintained 3 weeks after intervention termination.

Out of Seat

The percentage of intervals of OOS behaviors across participants is shown in Figure 3. A functional relation was established between the implementation of SIA intervention and OOS behaviors during WG instruction. Out-of-seat levels consistently decreased across all participants when SIA intervention was implemented to levels lower than baseline engagement levels; descriptions of data patterns by participants are described in the following sections.

Joy's OOS behaviors displayed an increasing trend during baseline, ranging from 20% to 42% of intervals. A decreasing trend at the onset of the intervention was observed between Sessions 5 and 14; behaviors increased in Session 15 due to health issues, then returned to lower

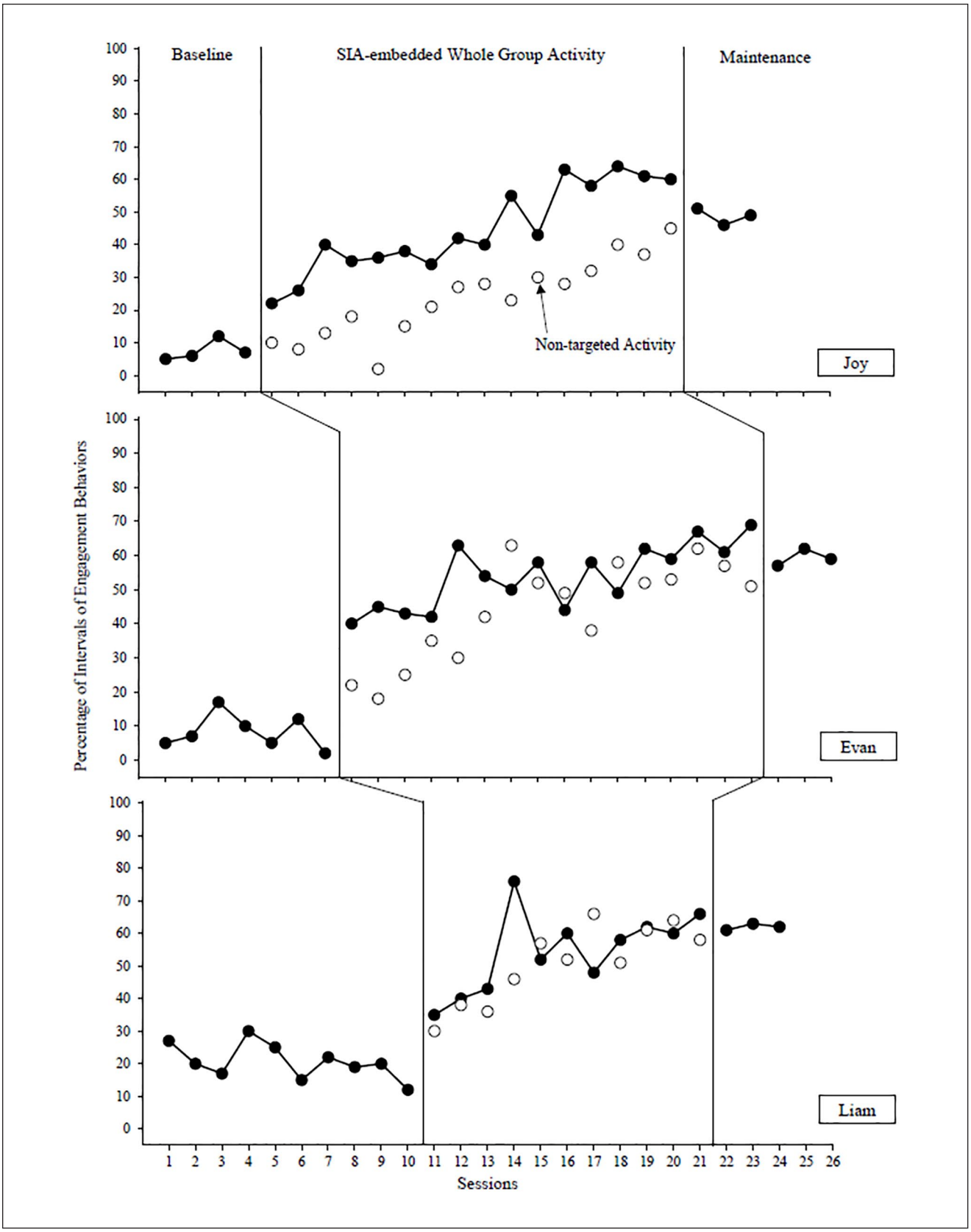


Figure 2. Children's Percentage of Intervals of Engagement Behaviors
Note. SIA = special interest area.

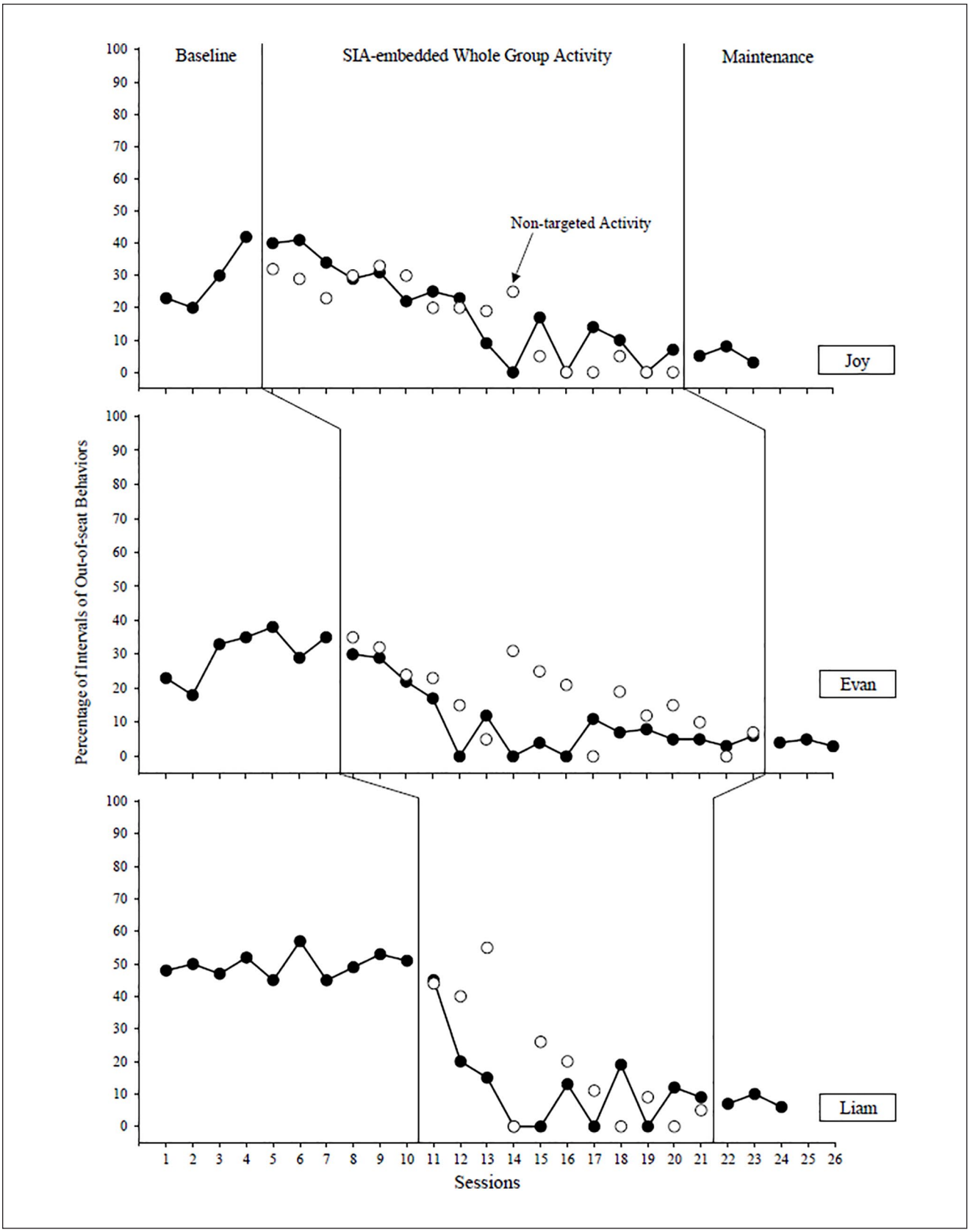


Figure 3. Children's Percentage of Intervals of Out-of-Seat Behaviors
Note. SIA = special interest area.

Table 3. Social Validity Scaling Questionnaire.

Question	Joy's teacher	Evan's teacher	Liam's teacher	Average score (range)
1. Embedding SIA in the activity plan was easy.	5	5	4	4.7 (4–5)
2. Implementing a modified activity plan was easy.	5	5	5	5.0
3. I observed an increase in the engagement behaviors of the child after the implementation of this intervention.	4	5	4	4.3 (4–5)
4. I observed a decrease in the out-of-seat behaviors of the child after the implementation of this intervention.	5	5	4	4.7 (4–5)
5. Using this strategy positively influenced the activity instruction in the inclusive classroom.	5	4	5	4.7 (4–5)
6. I would like to use this strategy in the future.	5	5	5	5.0
7. I would recommend this strategy to other teachers in the inclusive classroom.	5	5	5	5.0

Note. The following rating scale was used for the questionnaire: 1 = *Strongly disagree*, 2 = *Disagree*, 3 = *Neutral*, 4 = *Agree*, 5 = *Strongly agree*. SIA = special interest area.

levels. Overall, she exhibited a low level (0%–33% of the intervals) of behavior across intervention sessions. The level of behavior during the non-targeted probe sessions was similar to intervention sessions. Evan's OOS behaviors displayed an increasing trend during baseline, ranging from 18% to 38% of intervals. A decreasing trend at the onset of intervention was observed between Sessions 8 and 12, and he maintained a low level of behavior for the remaining sessions (around 10% of intervals). The levels of behavior displayed during non-targeted probe sessions matched with those during the intervention sessions. Liam's OOS behaviors displayed an increasing trend during baseline, ranging from 45% to 57% of intervals. His level of behavior gradually decreased and has shown a decreasing trend with minimal changes across intervention sessions. Data in both SIA-embedded and non-targeted conditions for all three children had a decreasing trend; the low levels of behaviors were maintained 3 weeks after intervention termination.

Social Validity

The average score across the items of the 5-point Likert-type rating scale questionnaire was 4.8 ($n = 3$), indicating the intervention was meaningful, acceptable, and feasible (see Table 3). According to responses to the open-ended questionnaires, the general educators' perspectives on using SIA-embedded instruction were (a) adding a few elements (e.g., SIA images) to an existing activity could change children's behaviors, (b) planning and implementing the intervention without collaborating with the special educator can be challenging, (c) they would like to expand the use of this strategy across other daily routines, and (d) seeing peers without disabilities showing greater interest on children with ASD as they witness their strengths and ability to participate in activities during the intervention was surprising. However, one mentioned planning, implementing, and evaluating every session was burdensome and difficult at first.

Discussion

This study examined the effects of embedding SIAs in WG instruction on the engagement and OOS behaviors of young children with ASD. Embedding SIAs in WG instruction increased the engagement and decreased OOS behaviors of children. For all children, the effects of the intervention were maintained 3 weeks after the interventions were terminated. Overall, this study provides support for the use of SIAs during WG instruction for children with ASD in early childhood inclusive settings.

This study presents several implications for practice. First, incorporating SIAs in WG instruction may result in an immediate improvement in engagement and OOS behaviors of children with ASD. This intervention was implemented within an ongoing daily routine as a preventative approach with low-intensity effort. In early childhood settings, accommodations, modifications, and adaptations are used to create an environment to promote learning and engagement for all children (Barton & Smith, 2015), rather than developing a separate program or changing the steps or contents of classroom activities for the individual who needs additional support. In the current study, all three participants had an increase in engagement and a decrease in their OOS behavior while maintaining the given curriculum and continuing the patterns of daily routines. Moreover, the results of the non-targeted probe condition data suggest perhaps the SIA-embedded strategy supported the children in learning the instructional routine. Thus, incorporating SIA during the beginning of the school year for children who have limited learning experiences may provide a pathway to teach classroom routines and orient children to instruction. Second, general educators

implemented the SIA-embedded instructional intervention with fidelity and reported the procedures were feasible and acceptable. Developing a modified activity plan took only 15 to 20 min of discussions between the general and special educators. In addition, identifying SIAs for each participant child involved indirect assessments by educators and caregivers, which took less time than systematic assessment of preferred items (Hagopian et al., 2004). This implies that it may not be necessary for educators to use direct assessments; rather, they could incorporate SIAs in activities as initial low-intensity support. Nevertheless, one of the general educators reported through the social validity questionnaire the first few weeks of this study were challenging and burdensome for her to plan, implement, and evaluate because she never received any type of related training or had professional development opportunities. This finding suggests that educator preparation or professional development programs may include opportunities for the general educators to be equipped to use relatively simple intervention strategies such as SIA-embedded interventions to effectively and efficiently promote engagement and OOS behaviors of children with ASD included in their classrooms. Finally, the general educators noted the collaborative relationship contributed to the high feasibility of the SIA-embedded interventions. This highlights the importance of collaboration, which aligns with critical components of creating high-quality inclusive settings emphasized in the Division for Early Childhood (DEC) Recommended Practice (DEC, 2014). While lack of time has been reported by educators as a barrier to communication and collaboration (Kritikos & Birnbaum, 2003), in the current study, different means of communication (e.g., formal/informal meetings, messaging programs, emails, and so on) were used to efficiently plan and discuss the instruction.

Future research can expand the use of SIAs by investigating the utility and possible generalization of improvements in engagement across activities and participants. Although there is limited research focused on using individualized, antecedent interventions to improve the engagement of children with ASD during WG instruction (e.g., Zimmerman et al., 2019), SIAs present a promising low-intensity strategy for improving access to instruction without extensive assessment of CB. However, the process by which the SIAs operated as tools to improve student engagement is unclear. All children in this study were learners with limited school histories, so it could be the SIAs that cued children new to the school to activities by incorporating their interests, and thus SIAs functioned as stimuli to support children accessing routines and activities. This process may be supported by the increases in engagement when SIAs were not present during non-targeted activities across all participants, a finding incongruent to the theoretical necessity of SIAs to be present for behavior change to occur. Future research should investigate if SIAs can be used to cue children new to classroom or community environments as a tool to support children's acquisition of routines in novel learning environments.

Furthermore, if SIAs operate as a process to cue children to access ongoing activities and routines to support their ability to learn how to learn (e.g., looking at the instructor or materials is important to acquire new information), then incorporating all children's interests (i.e., Instruction1; DEC, 2014; National Association for the Education of Young Children [NAEYC], 2009), not just those with ASD, may be beneficial for educators of young children new to schooling. Perhaps young children with ASD or similar language and behavioral support needs may be able to acquire more skills in learning environments if SIAs are strategically incorporated into instruction and routines initially before educators wait for children to need more intensive interventions or supports. However, the process by which SIAs can be used as a universal support, including the selection of SIAs and effectively embedding them in activities and routines, needs to be explored further for young children with initial school experiences before these claims can be made. Lastly, SIAs may have served as intermittent reinforcers for engagement behaviors. Given that SIAs were sometimes present or absent during the instruction, it could be that children were unable to predict the availability of SIAs and thus attended to instruction to potentially access the SIA. The increases in engagement and decreases in OOS behaviors across targeted and non-targeted WG instructions support this notion, although the functionality of SIAs as intermittent reinforcers was beyond the scope of this study. Not all SIAs function as reinforcement; thus, future research should explicitly evaluate if SIAs function as reinforcers for children with ASD and other language/behavioral support needs. Furthermore, future research should explore if intermittent access to SIAs can serve as a scheduled reinforcement to increase engagement and decrease OOS behaviors across the school day.

This study should be considered in light of several limitations. First, we utilized the educators' observation and caregivers' reports to identify SIAs for each participant instead of conducting a direct preference assessment procedure. Given the lack of correlations between the outcomes of direct and indirect preference assessments (Green et al., 1991), utilizing both measures is recommended as the most efficacious method to identify the SIAs (Hagopian et al., 2004). Additional research is needed to investigate whether the SIAs identified via indirect assessment can consistently function as the stimulus embedded in the instruction to improve engagement and OOS behaviors. Second, the social validity data were collected from general educators by special educators (i.e., the first author), which may be possible that the relationship with the first author affected their answers. Therefore, the result of social validity should

be interpreted in consideration of these circumstances. Moreover, although the focus of this study was to examine the behavioral change of children with ASD, future studies may further measure children's learning and their peers' behavioral outcomes. Measuring learning outcomes may serve as evidence demonstrating the effects of improving engagement on the child's development. Examining peers' behavioral changes helps to understand the effect of implementing SIA-embedded instruction in general education classrooms on promoting engagement in all children with or without ASD. Finally, the most notable limitation of the study was the use of partial-interval recording procedures to estimate engagement behaviors. Engagement behaviors are duration-based, and thus, the estimation system with the least amount of measurement error is momentary time sampling (Lane & Ledford, 2014). Given both engagement and OOS behaviors were measured simultaneously in this study, the authors decided to use a single estimation system given the limitations in research personnel and resources. Reported levels of engagement may be overestimated given partial interval recording was used (Lane & Ledford, 2014), although these potential overestimations impacted all conditions and participants. Future research estimating engagement should use momentary time sampling procedures rather than partial interval procedures to reduce the likelihood of measurement error.

Authors' Note

Gospel Y. Kim is now affiliated to UC Daivs MIND Institute, Sacramento, CA, USA; Jose R. Martinez is now affiliated to EIs for Autism Foundation, Jupiter, FL, USA

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Gospel Y. Kim https://orcid.org/0000-0002-3752-9261
Kathleen N. Tuck https://orcid.org/0000-0003-0271-0965

References

Adamson, L. B., Bakeman, R., & Deckner, D. F. (2004). The development of symbol-infused joint engagement. *Child Development*, *75*(4), 1171–1187. https://doi.org/10.1111/j.1467-8624.2004.00732.x

American Psychiatric Association. (2013). *Diagnostic and Statistical Manual of Mental Disorders: Diagnostic and Statistical Manual of Mental Disorders* (5th ed.).

Barton, E. E., & Smith, B. J. (2015). Advancing high-quality preschool inclusion: A discussion and recommendations for the field. *Topics in Early Childhood Special Education*, *35*(2), 69–78. https://doi.org/10.1177/0271121415583048

Bennett, K., Reichow, B., & Wolery, M. (2011). Effects of structured teaching on the behavior of young children with disabilities. *Focus on Autism and Other Developmental Disabilities*, *26*(3), 143–152. https://doi.org/10.1177/1088357611405040

Bishop, S. L., Hus, V., Duncan, A., Huerta, M., Gotham, K., Pickles, A., & Lord, C. (2013). Subcategories of restricted and repetitive behaviors in children with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *43*(6), 1287–1297. https://doi.org/10.1007/s10803-012-1671-0

Boyd, B. A., Conroy, M. A., Mancil, G. R., Nakao, T., & Alter, P. J. (2007). Effects of circumscribed interests on the social behaviors of children with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *37*(8), 1550–1561. https://doi.org/10.1007/s10803-006-0286-8

Boyd, B. A., McDonough, S. G., Rupp, B., Khan, F., & Bodfish, J. W. (2011). Effects of a family-implemented treatment on the repetitive behaviors of children with autism. *Journal of Autism and Developmental Disorders*, *41*(10), 1330–1341. https://doi.org/10.1007/s10803-010-1156-y

Chazin, K. T., Ledford, J. R., Barton, E. E., & Osborne, K. C. (2018). The effects of antecedent exercise on engagement during large group activities for young children. *Remedial and Special Education*, *39*(3), 158–170. https://doi.org/10.1177/0741932517716899

Crosland, K., & Dunlap, G. (2012). Effective strategies for the inclusion of children with autism in general education classrooms. *Behavior Modification*, *36*(3), 251–269. https://doi.org/10.1177/0145445512442682

DiCarlo, C. F., Pierce, S. H., Baumgartner, J., Harris, M. E., & Ota, C. (2012). Whole-group instruction practices and children's attention: A preliminary report. *Journal of Research in Childhood Education*, *26*(2), 154–168. https://doi.org/10.1080/02568543.2012.657744

Division for Early Childhood. (2014). *DEC recommended practices in early intervention/early childhood special education 2014*. http://www.dec-sped.org/recommendedpractices

El Zein, F., Solis, M., Vaughn, S., & McCulley, L. (2014). Reading comprehension interventions for students with autism spectrum disorders: A synthesis of research. *Journal of Autism and Developmental Disorders*, *44*(6), 1303–1322. https://doi.org/10.1007/s10803-013-1989-2

Fisher, W. W., Pawich, T. L., Dickes, N., Paden, A. R., & Toussaint, K. (2014). Increasing the saliency of behavior-consequence relations for children with autism who exhibit persistent errors. *Journal of Applied Behavior Analysis*, *47*(4), 738–748. https://doi.org/10.1002/jaba.172

Fleury, V. P., & Schwartz, I. S. (2017). A modified dialogic reading intervention for preschool children with autism spectrum disorder. *Topics in Early Childhood Special Education*, *37*(1), 16–28. https://doi.org/10.1177/0271121416637597

Gast, D. L., & Spriggs, A. D. (2014). Visual analysis of graphic data. In J. R. Ledford & D. L. Gast (Eds.), *Single case research methodology* (pp. 176–210). Routledge. https://doi.org/10.4324/9780203877937-9

Gilliam, W. S. (2005). *Prekindergarteners left behind: Expulsion rates in state prekindergarten systems*. Foundation for Child Development.

Green, C. W., Reid, D. H., Canipe, V. S., & Gardner, S. M. (1991). A comprehensive evaluation of reinforcer identification processes for persons with profound multiple handicaps. *Journal of Applied Behavior Analysis*, *24*(3), 537–552. https://doi.org/10.1901/jaba.1991.24-537

Gunn, K. C., & Delafield-Butt, J. T. (2016). Teaching children with autism spectrum disorder with restricted interests: A review of evidence for best practice. *Review of Educational Research*, *86*(2), 408–430. https://doi.org/10.3102/0034654315604027

Hagopian, L. P., Long, E. S., & Rush, K. S. (2004). Preference assessment procedures for individuals with developmental disabilities. *Behavior Modification*, *28*(5), 668–677. https://doi.org/10.1177/0145445503259836

Hemmeter, M. L., & Conroy, M. A. (2018). Advancement of evidence-based programs for young children with social and emotional learning difficulties. *School Mental Health*, *10*, 199–201. https://doi.org/10.1007/s12310-018-9275-2

Hemmeter, M. L., Snyder, P., Kinder, K., & Artman, K. (2011). Impact of performance feedback delivered via electronic mail on preschool teachers' use of descriptive praise. *Early Childhood Research Quarterly*, *26*(1), 96–109. https://doi.org/10.1016/j.ecresq.2010.05.004

Keen, D. (2009). Engagement of children with autism in learning. *Australasian Journal of Special Education*, *33*(2), 130–140. https://doi.org/10.1375/ajse.33.2.130

Kiely, B., Migdal, T. R., Vettam, S., & Adesman, A. (2016). Prevalence and correlates of elopement in a nationally representative sample of children with developmental disabilities in the United States. *PLOS ONE*, *11*(2), Article e0148337. https://doi.org/10.1371/journal.pone.0148337

Kishida, Y., & Kemp, C. (2006). Measuring child engagement in inclusive early childhood settings: Implications for practice. *Australian Journal of Early Childhood*, *31*(2), 14–19. https://doi.org/10.1177/183693910603100204

Kodak, T., Paden, A., & Dickes, N. (2012). Training and generalization of peer-directed mands with non-vocal children with autism. *The Analysis of Verbal Behavior*, *28*(1), 119–124. https://doi.org/10.1007/BF03393112

Koegel, R., Kim, S., Koegel, L., & Schwartzman, B. (2013). Improving socialization for high school students with ASD by using their preferred interests. *Journal of Autism and Developmental Disorders*, *43*(9), 2121–2134. https://doi.org/10.1007/s10803-013-1765-3

Korea Ministry of Education. (2013). *Preschool curriculum for age 3 to 5*. Ministry of Education.

Kritikos, E. P., & Birnbaum, B. (2003). General education and special education teachers' beliefs regarding collaboration. *Learning Disabilities: A Multidisciplinary Journal*, *12*(3), 93–100.

Lane, J. D., & Ledford, J. R. (2014). Using interval-based systems to measure behavior in early childhood special education and early intervention. *Topics in Early Childhood Special Education*, *34*(2), 83–93. https://doi.org/10.1177/0271121414524063

Ledford, J. R., & Gast, D. L. (Eds.). (2018). *Single case research methodology* (3rd ed.). Routledge. https://doi.org/10.4324/9781315150666

Mancil, G. R., & Pearl, C. E. (2008). Restricted interests as motivators: Improving academic engagement and outcomes of children on the autism spectrum. *TEACHING Exceptional Children Plus*, *4*(6), n6.

McComas, J., Hoch, H., Paone, D., & El-Roy, D. (2000). Escape behavior during academic tasks: A preliminary analysis of idiosyncratic establishing operations. *Journal of Applied Behavior Analysis*, *33*(4), 479–493. https://doi.org/10.1901/jaba.2000.33-479

McWilliam, R. A., Trivette, C. M., & Dunst, C. J. (1985). Behavior engagement as a measure of the efficacy of early intervention. *Analysis and Intervention in Developmental Disabilities*, *5*(1–2), 59–71. https://doi.org/10.1016/s0270-4684(85)80006-9

Morris, K. R., & Horner, R. H. (2016). Positive behavior support. In W. Sailor, G. Dunlap, G. Sugai, & R. Horner (Eds.), *Handbook of evidence-based practices in intellectual and developmental disabilities* (pp. 415–441). Springer. https://doi.org/10.1007/978-3-319-26583-4_15

National Association for the Education of Young Children. (2009). *NAEYC position statement on developmentally appropriate practice in early childhood programs serving children from birth to age 8*.

Ninci, J., Neely, L. C., Burke, M. D., Rispoli, M., Vannest, K. J., Willson, V., & Ulloa, P. (2020). A meta-analysis of single-case research on embedding interests of individuals with autism spectrum disorder. *Remedial and Special Education*, *41*(5), 296–308.

Ogundele, M. O. (2018). Behavioural and emotional disorders in childhood: A brief overview for pediatricians. *World Journal of Clinical Pediatrics*, *7*(1), 9–26. https://doi.org/10.5409/wjcp.v7.i1.9

Powell, D. R., Burchinal, M. R., File, N., & Kontos, S. (2008). An eco-behavioral analysis of children's engagement in urban public school preschool classrooms. *Early Childhood Research Quarterly*, *23*(1), 108–123. https://doi.org/10.1016/j.ecresq.2007.04.001

Raspa, M. J., McWilliam, R. A., & Maher Ridley, S. (2001). Child care quality and children's engagement. *Early Education and Development*, *12*(2), 209–224. https://doi.org/10.1207/s15566935eed1202_3

Ridley, S. M., McWilliam, R. A., & Oates, C. S. (2000). Observed engagement as an indicator of child care program quality. *Early Education and Development*, *11*(2), 133–146. https://doi.org/10.1207/s15566935eed1102_1

Rispoli, M., Lang, R., Neely, L., Camargo, S., Hutchins, N., Davenport, K., & Goodwyn, F. (2013). A comparison of within-and across-activity choices for reducing challenging behavior in children with autism spectrum disorders. *Journal of Behavioral Education*, *22*(1), 66–83. https://doi.org/10.1007/s10864-012-9164-y

Rogers, S. J., & Dawson, G. (2020). *Early Start Denver Model for young children with autism: Promoting language, learning, and engagement*. Guilford Publications.

Smith, B., & Fox, L. (2003). *Systems of service delivery: A synthesis of evidence relevant to young children at risk of or*

who have challenging behavior. Center for Evidence-based Practice: Young Children with Challenging Behavior.

Steinbrenner, J. R., Hume, K., Odom, S. L., Morin, K. L., Nowell, S. W., Tomaszewski, B., Szendrey, S., McIntyre, N. S., Yucesoy-Ozkan, S., & Savage, M. N. (2020). *Evidence-based practices for children, youth, and young adults with Autism.* National Clearinghouse on Autism Evidence and Practice Review Team, Frank Porter Graham Child Development Institute, The University of North Carolina at Chapel Hill. https://ncaep.fpg.unc.edu/sites/ncaep.fpg.unc.edu/fles/imce/documents/EBP%20Report%202020.pdf

Willford, A. P., Maier, M. F., Downer, J. T., Pianta, R. C., & Howes, C. (2013). Understanding how children's engagement and teachers' interactions combine to predict school readiness. *Journal of Applied Developmental Psychology*, *34*(6), 299–309. https://doi.org/10.1016/j.appdev.2013.05.002

Zimmerman, K. N., Ledford, J. R., & Severini, K. E. (2019). Brief report: The effects of a weighted blanket on engagement for a student with ASD. *Focus on Autism and Other Developmental Disabilities*, *34*(1), 15–19. https://doi.org/10.1177/1088357618794911